

Program 6: Deploy an Automation build pipeline using Dockerhub

Step 1: mkdir program6

cd program6

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops$ cd program6
```

Step 2: Create a Dockerfile

nano Dockerfile

Step 3: Create a PAT token

PAT for Dockerhub

PAT for Github

Copy the passwords and save it in a file

Step 4: initialize the npm packages

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ npm init -y
Wrote to /home/1rv24mc040-haripriya/Devops/program6/package.json:
```

```
{
  "name": "program6",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

server.js

```
const express = require("express");
const app = express();

app.get("/", (req, res) => {
  res.send("Hello from Docker automated pipeline!");
});

app.listen(3000, () => console.log("Server running on port 3000"));
```

Step 5: Login to the dockerhub using the PAT

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ docker login -u haripriya1701
```

 Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit <https://app.docker.com/settings>

Password:

WARNING! Your credentials are stored unencrypted in '/home/1rv24mc040-haripriya/.docker/config.json'.

Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded

Step 6: Build the image

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ docker build -t program6 .
[+] Building 315.7s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 285B                                0.0s
=> [internal] load metadata for docker.io/library/node:18         2.8s
=> [auth] library/node:pull token for registry-1.docker.io        0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d 311.5s
=> => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d39 0.0s
=> => sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a 6.41kB / 6.41kB 0.0s
=> => sha256:eb29363371ee2859fad6a3c5af88d4abc6ff7d399ad 2.49kB / 2.49kB 0.0s
=> => sha256:37927ed901b1b2608b72796c6881bf645480268e 24.02MB / 24.02MB 41.8s
=> => sha256:79b2f47ad4443652b9b5cc81a95ede249fd97631 64.40MB / 64.40MB 55.4s
```

Step 7: Tag the image for dockerhub

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ docker tag program6 haripriya1701/program6:latest
```

Step 8: Push the image to the dockerhub

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ docker push haripriya1701/program6:latest
The push refers to repository [docker.io/haripriya1701/program6]
97c0489dc73d: Pushed
d9e96791b291: Pushed
879d814895cf: Pushed
736b2d292483: Pushed
d2a991bcab4d: Mounted from library/node
b624aa2d5ea2: Mounted from library/node
d399c9dc306f: Mounted from library/node
84f9fa179c1b: Mounted from library/node
ce84ba212e49: Mounted from library/node
e4dc8cd9ecc8: Mounted from library/node
6428cc293366: Mounted from library/node
2f7436e79a0b: Mounted from library/node
latest: digest: sha256:3332f9c7b0733f2a0f769f2cd04f8400b1d11037080dc3c5dc7da590321b0ef7 size: 2831
```

Step 9: Login to the ghcr using the PAT

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ echo ghp_WLaMpXsi7gx9oRhcoYdI6HEf0aeaw63nGEnx | docker login ghcr.io -u Haripriyaa17 --password-stdin

WARNING! Your credentials are stored unencrypted in '/home/1rv24mc040-haripriya/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
```

Step 10: Tag the image for the GHCR

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ docker tag program6 ghcr.io/haripriyaa17/program6:latest
```

Step 11: Push the image to the GHCR

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ docker push ghcr.io/haripriyaa17/program6:latest
The push refers to repository [ghcr.io/haripriyaa17/program6]
97c0489dc73d: Pushed
d9e96791b291: Pushed
879d814895cf: Pushed
736b2d292483: Pushed
d2a991bcab4d: Pushed
b624aa2d5ea2: Pushed
d399c9dc306f: Pushed
84f9fa179c1b: Pushed
ce84ba212e49: Pushed
e4dc8cd9ecc8: Pushed
6428cc293366: Pushed
2f7436e79a0b: Pushed
latest: digest: sha256:60456f382057407af2a054df8bdfc7ac8b93eb144480a95e2038549031423c65 size: 2831
```

Step 12: Pushing the files to the Github repo

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1rv24mc040-haripriya/Devops/program6/.git/
```

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ git add .
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ git commit -m "Initial commit"
[master (root-commit) ba4f038] Initial commit
 4 files changed, 53 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 package-lock.json
 create mode 100644 package.json
 create mode 100644 server.js
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ git remote add origin https://github.com/haripriyaa17/program6.git
```

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program6$ git push -u origin main
Username for 'https://github.com': Haripriyaa17
Password for 'https://Haripriyaa17@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 950 bytes | 950.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
remote: This repository moved. Please use the new location:
remote:   https://github.com/Haripriyaa17/program6.git
To https://github.com/haripriyaa17/program6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Step 13: Check the following for the pushed files

Dockerhub

The screenshot shows the Docker Hub interface for a repository named `haripriya1701/program6`. The repository is public and has been pushed about 1 hour ago. The repository size is 424.3 MB, and it has 0 stars and 37 downloads. The interface includes a sidebar with navigation options like Repositories, Hardened Images, Collaborations, Settings, Default privacy, Notifications, Billing, Usage, Pulls, and Storage. The main content area shows the repository details, including a table of tags. The table has columns for Tag, OS, Type, Pulled, and Pushed. The only tag listed is `latest`, which is an image type, pushed less than 1 day ago, and pulled about 1 hour ago. There is also a section for Docker commands, showing the command `docker push haripriya1701/program6:tagname`. A sidebar on the right promotes Docker Build Cloud.

Tag	OS	Type	Pulled	Pushed
<code>latest</code>		Image	less than 1 day	about 1 hour

Github repo

The screenshot shows the GitHub interface for a repository named `Haripriya17/program6`. The repository is public and has 1 branch and 0 tags. The repository contains several files: `Dockerfile`, `package-lock.json`, `package.json`, and `server.js`. The repository was created by Haripriya17 and has 1 commit. The repository is currently on the `main` branch. The repository has no description, website, or topics provided. The repository has 0 stars, 0 watching, and 0 forks. The repository has no releases published. The repository has no packages published.

GHCR package

The screenshot shows the GitHub Container Registry (GHCR) interface for a package named `program6`. The package is public and has the latest version published. The package was published 28 minutes ago and has 0 total downloads. The package has no settings. The package is installed from the command line using the command `$ docker pull gcr.io/haripriya17/program6:latest`. The package has a recent tagged image version named `latest`, which was published 28 minutes ago and has a digest of `sha256:...`. The package has a view and manage all versions link.